

0 4 . 1

This question is about the flow of blood into and through the heart.

Add the numbers 1 to 6 to **Table 2** to give the order of structures through which blood will pass as it enters the heart and flows through the left ventricle.

Use each number only once. Number **4** has been done for you.

[2 marks]

Table 2

Aorta		Pulmonary vein	
Left atrioventricular valve		Left semi-lunar valve	
Right atrioventricular valve		Vena cava	
Left atrium		Left ventricle	4
Right atrium		Right ventricle	
Pulmonary artery		Right semi-lunar valve	



Question 4 continues on the next page

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ANSWER IN THE SPACES PROVIDED**

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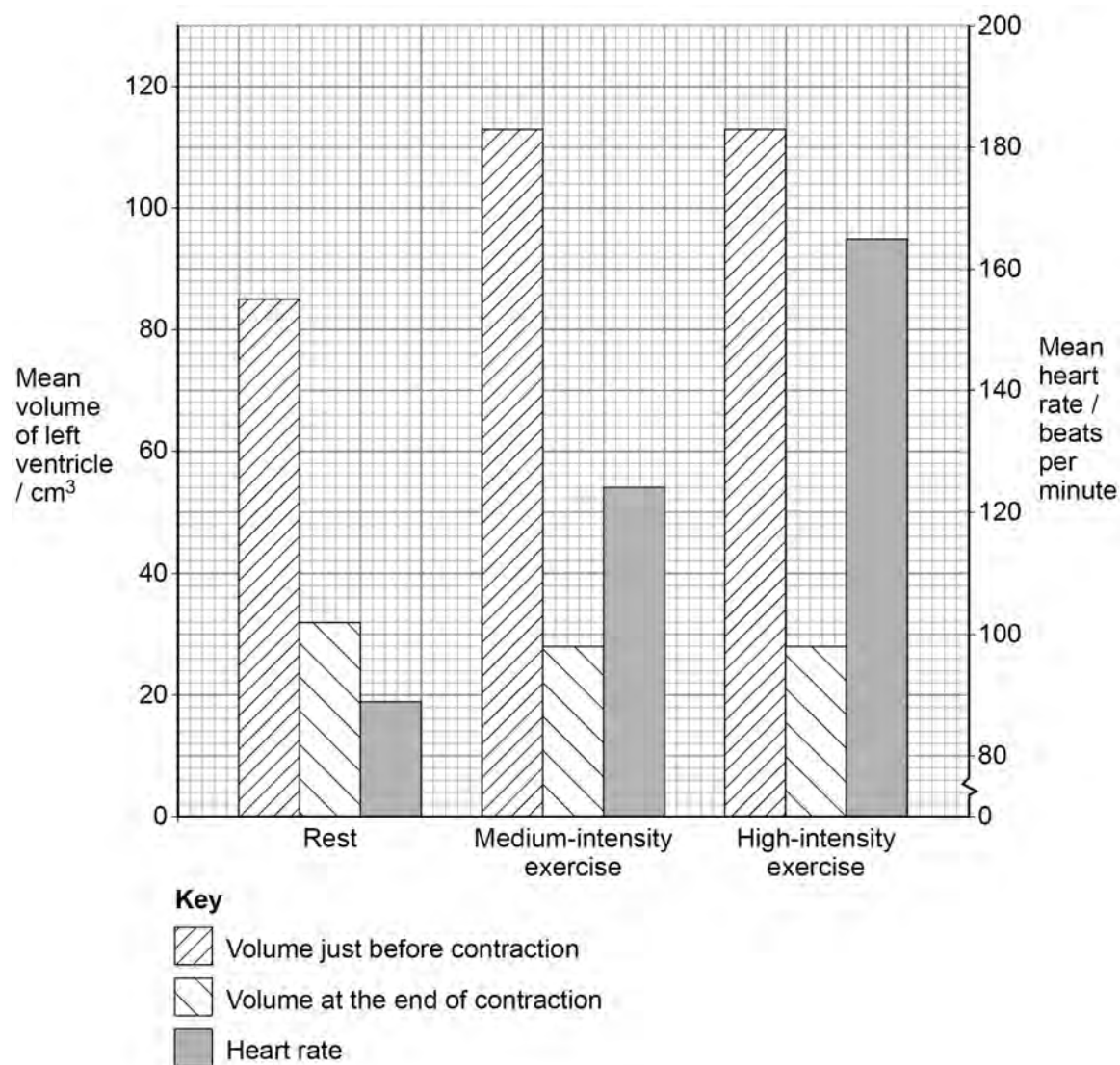


0 4 . 2

Scientists investigated the heart activity of humans at rest, during medium-intensity exercise and during high-intensity exercise.

Figure 2 shows the scientists' results.

Figure 2



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Stroke volume = volume of blood leaving a ventricle with each contraction

Cardiac output = stroke volume $\times$ heart rate

Use all the information to describe what causes the increase in cardiac output:

- from rest to medium-intensity exercise
- from medium-intensity exercise to high-intensity exercise.

You do **not** need to calculate cardiac output to answer this question.

[2 marks]

Rest to medium-intensity exercise _____

Medium-intensity exercise to high-intensity exercise _____

0 4 . 3

Name the type of blood vessel that controls blood flow to muscles **and** explain how these blood vessels change blood flow during exercise.

[3 marks]

Name of blood vessel _____

Explanation _____

7

Turn over for the next question

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